# Research Statement Draft

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Research Interests: AI Safety, Interpretable AI, Autonomous Intelligent Systems, Generative AI, Multi-Agent Systems

## Background

My background is in Artificial Intelligence, Machine Learning, and intelligent autonomous systems, with research experience spanning interpretable AI, generative AI, distributed sensing systems, and real-world machine learning applications.

I completed my Master's degree in Computer Science with specialization in AI & Data Science at IIITDM Kancheepuram, India, including research work conducted at the University of Agder (UiA), Norway. During this period, I worked as a Visiting Researcher and later as a Research Assistant at CAIR (Center for Artificial Intelligence Research), UiA.

My Master's thesis focused on distributed FMCW radar point cloud clustering and fusion for multi-human intruder detection and tracking. The work involved designing autonomous perception pipelines for distributed radar systems operating under sparse and noisy real-world conditions. Through this work, I became increasingly interested in how intelligent systems reason under uncertainty, adapt to evolving environments, and make reliable decisions in safety-critical settings.

My research experiences across interpretable AI, autonomous systems, and generative modeling motivated my long-term interest in trustworthy and human-aligned AI systems.

## Current Research

Currently, I am working at CAIR, University of Agder, on generative probabilistic modeling using Tsetlin Machines (TM). The research explores how interpretable rule-based learning systems can be extended from discriminative learning toward generative AI frameworks.

Our work focuses on constructing generative probabilistic models using collections of Tsetlin Machine classifiers to approximate conditional and joint probability distributions for data generation. We investigate autoregressive and Gibbs-sampling-inspired approaches for generating image data while maintaining interpretability and uncertainty awareness.

One particularly interesting aspect of this research is studying how Tsetlin-based generative systems behave under uncertainty and extrapolation. Unlike many neural generative models that may hallucinate plausible but unsupported outputs, the probabilistic behavior of Tsetlin Machines may provide signals indicating when the model operates outside the learned data distribution. I find this direction especially exciting because it connects interpretable AI with trustworthy and controllable generative systems.

In parallel, I have also worked on projects involving human activity recognition, distributed sensing systems, radar-based perception, UAV classification, and deep learning for autonomous real-world environments.

## Research Interests

My primary research interest is AI safety and alignment for advanced autonomous and generative AI systems. I am particularly interested in building intelligent systems that remain interpretable, reliable, controllable, and aligned with human values as they become increasingly adaptive and autonomous.

My recent work on interpretable generative probabilistic modeling strengthened my interest in trustworthy generative AI, uncertainty-aware reasoning, and controllable intelligent systems. I am especially interested in understanding how interpretable and rule-based approaches can improve robustness, transparency, and safety in future AI systems and autonomous agents.
Some research directions I am particularly excited about include:

- Context-aware and memory-enabled AI agents,
- Interpretable and controllable generative AI,
- Long-term behavioral adaptation in autonomous systems,
- Evaluation of emergent capabilities and AI reliability,
- Cooperative and multi-agent systems,
- AI safety in open-ended and evolving environments,
- Mechanisms for detecting uncertainty, hallucination, and out-of-distribution behavior in generative systems.

I am especially motivated by research that combines foundational AI safety questions with practical real-world deployment challenges.

## COMPASS LAB

I am particularly excited about the COMPASS group because its research vision strongly aligns with my interests in trustworthy autonomous systems, interpretable AI, and AI safety for advanced intelligent agents.

I am especially inspired by the group's work on indirect prompt injection attacks, contextual reasoning, open-ended safety, and cooperative AI systems. The focus on proactively understanding and mitigating safety risks in advanced AI agents resonates deeply with the type of research problems I hope to work on in the future.

What attracts me most is the lab's balance between foundational research, real-world impact, and forward-looking thinking about future AI systems. I am also highly motivated by the interdisciplinary and collaborative AI research environment in Tübingen, as well as the opportunity to contribute to impactful research that addresses both technical and societal challenges in AI safety.

## Future Goals

My long-term goal is to contribute to the development of reliable, interpretable, and human-aligned AI systems that can safely operate in complex real-world environments.
Through a PhD, I hope to deepen my understanding of AI safety, autonomous reasoning systems, and trustworthy generative AI while developing rigorous research skills that enable me to contribute meaningfully to the broader AI safety community.

I am particularly interested in working on research that bridges foundational AI safety questions with practical deployment challenges, especially in adaptive and autonomous systems that continuously interact with humans and evolving environments.

Ultimately, I hope to contribute toward building advanced AI systems that are not only capable and intelligent, but also transparent, controllable, cooperative, and beneficial to society.